



Industrial Internship Report on " AUTOMOTIVE STORE E-Commerce Website for Automotive Parts & Accessories"

**Prepared by
[AAVUTI SANJAY]**

Executive Summary

This report rapid growth of digital technology has significantly transformed the way people purchase products and services. E-commerce platforms have become an essential part of modern business by providing customers with convenient, efficient, and reliable online shopping experiences. In the automotive industry, the demand for online purchasing of vehicle spare parts and accessories has increased considerably due to the growing number of vehicle owners and the need for quick access to genuine products. Recognizing this need, the project titled "Automotive Store – E-Commerce Website for Automotive Parts & Accessories" was developed as part of the Industrial Internship to provide an efficient online platform for browsing and purchasing automotive components.

This project was carried out during the six-week Industrial Internship organized by Upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt. Ltd. (UCT). The internship provided practical exposure to modern software development practices, enabling the application of academic knowledge in a real-world environment. The primary objective of the project was to design and develop a responsive web application capable of displaying automotive products in an organized manner while providing users with a simple and interactive shopping experience.

The Automotive Store application has been developed using Python as the programming language and the Flask framework for backend development. The frontend has been designed using HTML5, CSS3, and Bootstrap 5, ensuring an attractive and responsive user interface that functions effectively across desktop and mobile devices. SQLite has been used as the backend database for storing product information, while

Flask session management has been utilized to maintain shopping cart data throughout the user's browsing session.

The application is organized into multiple product categories such as Engine Parts, Brake System, Electrical Parts, Suspension Parts, Lighting, Filters, Cooling System, Tyres & Wheels, Interior Accessories, Exterior Accessories, Exhaust System, and Safety & Security. Each category contains relevant automotive products along with images, prices, and descriptions, enabling users to browse products conveniently.

One of the major functionalities of the application is the Shopping Cart Module, which allows customers to add products, update quantities, remove unwanted items, and view the total amount before proceeding to checkout. Although advanced features such as Wishlist, Admin Panel, and Checkout Module are currently under development, the existing application successfully demonstrates the core functionality of an e-commerce platform.

During the development of this project, several technical concepts were explored, including Flask routing, template rendering using Jinja2, Bootstrap-based responsive design, session management, SQLite database integration, debugging, and structured project organization. Various challenges encountered during implementation, such as image rendering, shopping cart quantity management, Flask session handling, and application debugging, were systematically resolved through continuous testing and iterative development.

The internship significantly contributed to enhancing practical knowledge in full-stack web development, problem-solving, debugging techniques, database management, and software project implementation. It also strengthened professional skills such as analytical thinking, documentation, communication, and project planning, all of which are essential for a successful software engineering career.

The Automotive Store project successfully demonstrates the practical application of modern web technologies in developing a user-friendly e-commerce solution for automotive parts and accessories. The project serves as a strong foundation for future enhancements, including user authentication, secure online payment integration, order management, inventory management, customer reviews, product search and filtering, and administrative functionalities.

Overall, the internship provided valuable industrial exposure and practical experience in software development, enabling the successful completion of a functional web application while improving both technical competencies and professional skills. The knowledge gained throughout this internship will serve as a strong foundation for future academic projects and professional opportunities in the software development industry.

CONTENT	PAGE NO
1.PREFACE	4
2.INTRODUCTION	6
3.PROBLEM STATEMENT	16
4.EXISTING AND PROPOSED SOLUTION	16
5.PROPOSED DESIGN	25
6.PERFORMANCE TEST	33
7.MY LEARNINGS	36
8.FUTURE SCOPE	37
9.CONCLUSION	38

1 Preface

Industrial Internship is an integral part of the Master of Computer Applications (MCA) curriculum, designed to provide students with practical exposure to real-world software development environments. It enables students to apply theoretical concepts acquired during classroom learning to practical industry-oriented projects while gaining hands-on experience with modern technologies, software development methodologies, and professional work practices.

As part of this internship, I had the opportunity to work on the project "Automotive Store – E-Commerce Website for Automotive Parts & Accessories" under the guidance of Upskill Campus, in collaboration with The IoT Academy and UniConverge Technologies Pvt. Ltd. (UCT). This project provided an excellent platform to understand the complete software development lifecycle, beginning from requirement analysis and system design to implementation, testing, and documentation.

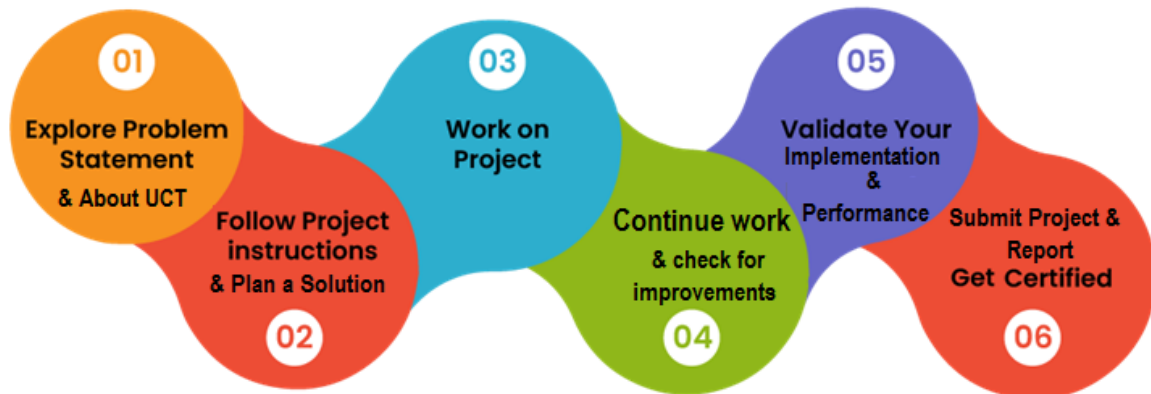
The primary objective of developing this application was to create a responsive and user-friendly e-commerce website that allows customers to explore automotive spare parts and accessories through an organized digital platform. The project involved designing intuitive user interfaces, implementing backend functionality using the Flask framework, integrating an SQLite database for product management, and developing shopping cart functionality using Flask session management.

Throughout the internship, I gained practical knowledge of various technologies including Python, Flask, HTML5, CSS3, Bootstrap, SQLite, Git, and GitHub. Apart from technical skills, the internship also enhanced my understanding of software engineering principles such as modular programming, debugging, database connectivity, responsive web design, project organization, documentation, and version control.

The internship helped me understand the importance of planning, teamwork, continuous learning, and systematic problem-solving in software development. Various implementation challenges, including database integration, routing, session management, responsive layout design, and debugging, were addressed through careful analysis and iterative improvements. These experiences significantly improved my confidence in developing real-world web applications.

This report documents the complete development process of the Automotive Store project, including the objectives, problem statement, existing system, proposed solution, technologies used, architecture, implementation methodology, testing procedures, challenges encountered, and future enhancement opportunities. The report has been prepared following the guidelines provided by the institution and reflects the knowledge and practical experience gained throughout the internship.

I sincerely hope that this report effectively demonstrates the learning outcomes achieved during the internship and highlights the successful application of academic knowledge in developing an industry-oriented software solution.



2 Introduction

About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform ()

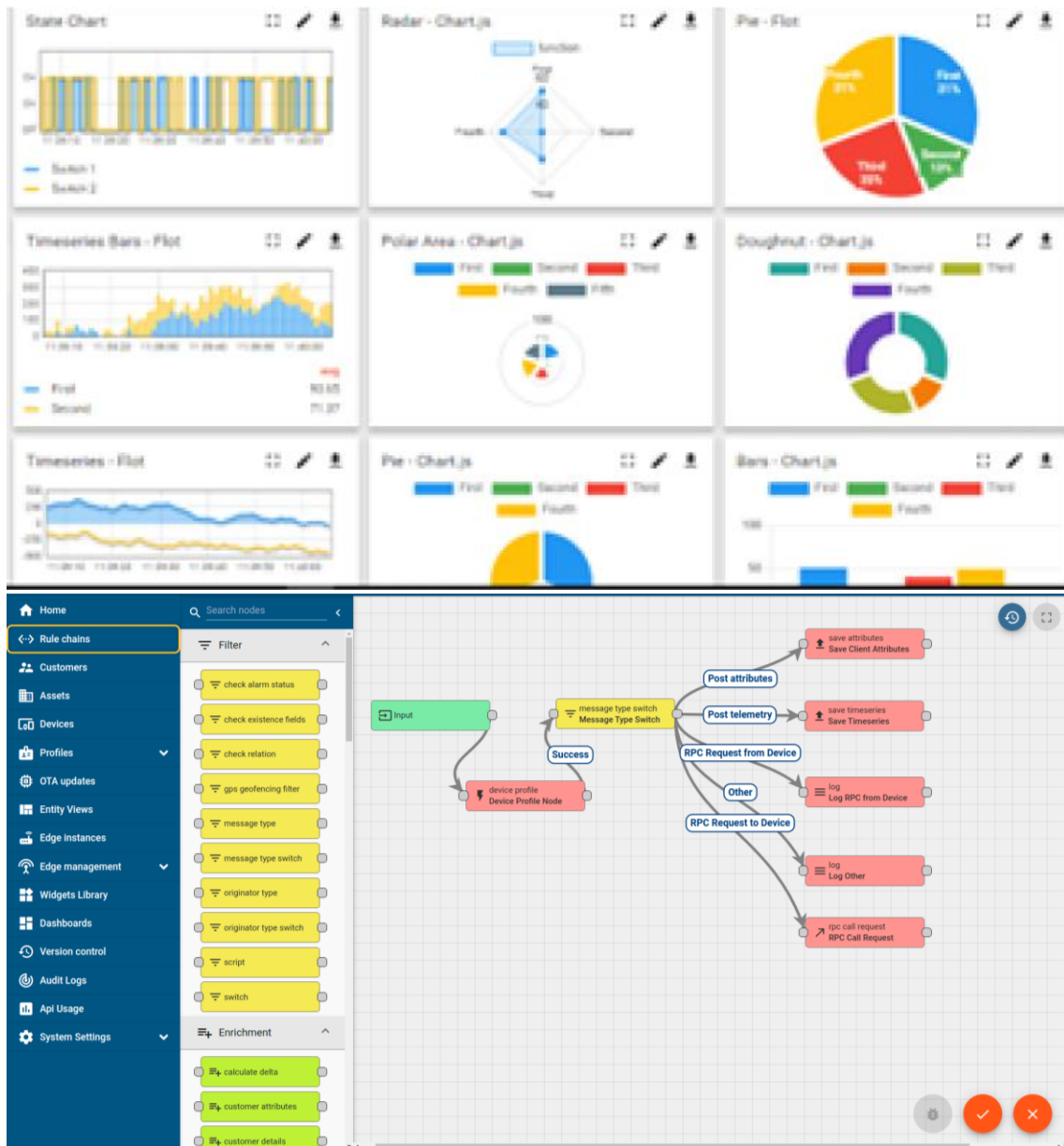
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification

- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

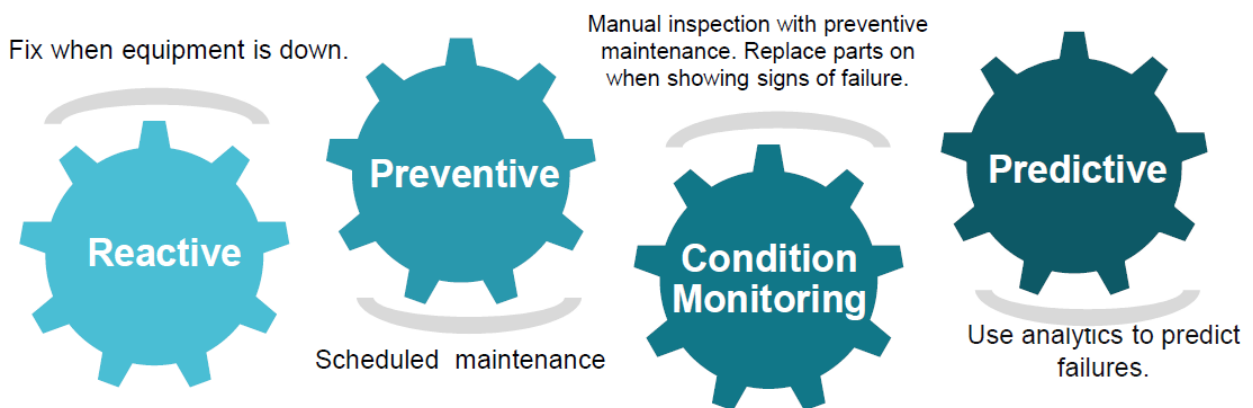
Its unique SaaS model helps users to save time, cost and money.

iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



About upskill Campus (USC)

Upskill Campus is one of India's leading online learning and career development platforms that focuses on bridging the gap between academic education and industry requirements. It provides students and young professionals with opportunities to develop practical skills through industry-oriented training programs, internships, certification courses, workshops, and project-based learning experiences.

The platform collaborates with reputed industries, technology partners, and educational institutions to provide learners with real-world exposure and hands-on experience. Through structured learning programs, Upskill Campus helps students understand modern technologies and industry practices while preparing them for successful careers in software development, data science, artificial intelligence, cybersecurity, cloud computing, and other emerging domains.

One of the major strengths of Upskill Campus is its focus on experiential learning. Instead of limiting students to theoretical concepts, it encourages learners to solve practical business problems by developing real-time projects under expert guidance. This approach enables students to improve analytical thinking, problem-solving abilities, teamwork, and communication skills.

During this Industrial Internship, Upskill Campus provided a structured learning environment where the complete software development lifecycle was followed—from understanding project requirements and planning to implementation, testing, debugging, and documentation. Regular assignments, practical exercises, and project reviews helped in continuously improving both technical and professional competencies.

The internship also encouraged self-learning and research by exposing participants to modern web development technologies and industry-standard development practices. Continuous mentorship and technical support played an important role in successfully completing the project.

Services Offered by Upskill Campus

Industrial Internship Programs

Skill Development Courses

Live Industry Projects

Career Guidance and Mentorship

Certification Programs

Placement Assistance

Technical Workshops

Industry-Oriented Training

Key Learning Outcomes.

The IoT Academy

The IoT Academy is a technology-focused training organization dedicated to providing industry-relevant education in emerging technologies. The academy specializes in delivering practical training programs that enable students and professionals to acquire technical skills aligned with current industry requirements.

The organization offers training in various domains including:

Internet of Things (IoT)

Python Programming

Full Stack Web Development

Artificial Intelligence

Machine Learning

Data Science

Cloud Computing

Cyber Security

The IoT Academy follows a practical learning approach where theoretical concepts are reinforced through hands-on implementation, assignments, and project development. Students receive continuous mentorship from experienced trainers who guide them through each stage of the learning process.

During the internship, The IoT Academy played an important role by providing technical guidance, learning materials, project support, and expert mentoring. The structured training sessions helped in understanding web application architecture, backend development using Flask, frontend design using Bootstrap, database integration using SQLite, and software debugging techniques.

The academy also emphasized professional development by encouraging learners to follow coding standards, write clean code, maintain proper project documentation, and adopt systematic software development methodologies.

The knowledge and practical experience gained through this training significantly contributed to the successful completion of the Automotive Store project and strengthened overall software development skills.

About UniConverge Technologies PVT . LTD. (UCT)

Uniconverge Technologies Pvt. Ltd. (UCT) is a technology-driven organization that focuses on providing innovative digital solutions and industry-oriented training programs. The organization

works across various domains of software development, digital transformation, cloud technologies, Internet of Things (IoT), data analytics, artificial intelligence, and enterprise software solutions.

UCT collaborates with educational institutions and learning platforms to bridge the gap between academic education and industrial requirements. Through internship programs and project-based learning, students gain valuable practical exposure to modern software engineering practices and real-world problem-solving techniques.

The organization encourages learners to understand the complete Software Development Life Cycle (SDLC), including:

Requirement Analysis

System Design

Development

Testing

Deployment

Maintenance

Documentation

During this internship, UCT provided industrial exposure that helped in understanding professional software development practices. The project was developed following modular programming principles, organized project structure, efficient debugging techniques, and systematic documentation.

Working under the internship program enabled practical implementation of various technologies including Python, Flask, HTML5, CSS3, Bootstrap, and SQLite. The internship also highlighted the importance of teamwork, planning, time management, quality assurance, and continuous learning.

The industrial exposure received through UCT has enhanced technical confidence and prepared the student for future opportunities in software development and information technology.

Objectives of this Internship program

The Industrial Internship was undertaken with the objective of applying academic knowledge to solve real-world software development problems while gaining practical industry experience. The internship focused on developing technical skills, improving professional competencies, and understanding the complete software development lifecycle.

The major objectives of the internship are as follows:

Technical Objectives

To understand the Software Development Life Cycle (SDLC).

To gain practical experience in Full Stack Web Development.

To develop a responsive web application using Python Flask.

To implement frontend technologies such as HTML5, CSS3, and Bootstrap.

To integrate SQLite as the backend database.

To implement session-based shopping cart functionality.

To understand routing and template rendering using Flask.

To improve debugging and error-handling skills.

Professional Objectives

To understand industrial software development practices.

To improve problem-solving and analytical thinking.

To enhance project planning and documentation skills.

To develop professional communication abilities.

To understand teamwork and collaborative development.

To improve software testing methodologies.

To gain confidence in implementing real-world projects.

Project Objectives

The primary objective of the project was to develop a responsive and user-friendly e-commerce website for automotive parts and accessories.

The application was designed to:

Display automotive products in multiple categories.

Provide detailed product information.

Allow users to add products to a shopping cart.

Update product quantities dynamically.

Calculate subtotal and total prices.

Maintain shopping cart information using Flask sessions.

Provide a responsive interface compatible with desktop and mobile devices.

Offer a scalable architecture for future enhancements.

The project also aims to support future implementation of advanced modules such as:

User Authentication

Wishlist

Admin Dashboard

Secure Checkout

Online Payment Gateway

Product Search and Filtering

Order History

Customer Reviews

Inventory Management

Overall, the internship successfully achieved its objectives by providing practical exposure to modern software development technologies and enhancing both technical knowledge and professional skills.

Reference

Python Software Foundation. *Python Documentation*. Available at: <https://docs.python.org/>

Pallets Projects. *Flask Documentation*. Available at: <https://flask.palletsprojects.com/>

Bootstrap Team. *Bootstrap 5 Documentation*. Available at: <https://getbootstrap.com/docs/>

SQLite Consortium. *SQLite Documentation*. Available at: <https://www.sqlite.org/docs.html>

Mozilla Developer Network (MDN). *HTML, CSS, and JavaScript Documentation*. Available at: <https://developer.mozilla.org/>

W3Schools. *HTML, CSS, Python, SQL and Bootstrap Tutorials*. Available at: <https://www.w3schools.com/>

Microsoft. *Visual Studio Code Documentation*. Available at: <https://code.visualstudio.com/docs>

Git Documentation. Available at: <https://git-scm.com/doc>

Glossary

Terms	Acronym
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
BOOTSTRAP	Front-end CSS Framework for Responsive Web Design
PYTHON	High-Level Programming Language
FLASK	Lightweight Python Web Framework
SQLite	Lightweight Relational Database Management System

HTTP	Hyper Text Transfer Protocol
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
IDE	Integrated Development Environment
VS CODE	Visual Studio Code
SDLC	Software Development Life Cycle
GIT	Distributed Version Control System
GIT HUB	Cloud Based Source Code Repository
CRUD	Create, Read, Update, Delete
RESPONSIVE DESIGN	Web Design That Adapts to Different Screen Sizes
E-COMMERCE	Electronic Commerce (Online Buying and Selling)
FRONTEND	User-facing Part of the Application
BACKEND	Server-side Logic and Database Processing
DATABASE	Structured Collection of Data
TEMPLATE	HTML Page Rendered by Flask Using Jinja2
JINJA 2	Template Engine Used by Flask

3 Problem Statement

The automotive industry offers thousands of spare parts and accessories that are traditionally sold through physical retail outlets. Customers often spend considerable time visiting multiple stores to locate suitable products, compare prices, and verify product availability. This process becomes even more difficult when specialized or less commonly available components are required.

Many traditional automotive stores still rely on manual inventory management and offline purchasing methods, resulting in inefficient customer service and limited accessibility. Customers are unable to determine product availability before visiting the store, leading to unnecessary travel and inconvenience.

The lack of centralized digital platforms also makes it difficult for customers to compare different automotive products efficiently. Manual purchasing processes increase the possibility of incorrect product selection, pricing inconsistencies, and delays in obtaining required components.

Furthermore, small automotive retailers often lack affordable digital solutions to showcase their products online. Without an effective web application, businesses are unable to reach a wider customer base and provide modern shopping experiences expected by today's consumers.

Therefore, there is a need for a simple, efficient, and responsive web application that enables users to browse automotive parts, organize products into categories, manage shopping carts, and improve the overall purchasing experience.

The Automotive Store project has been developed to address these challenges by providing a modern web-based e-commerce platform for automotive parts and accessories.

4 Existing and Proposed solution

Before developing the Automotive Store application, the purchasing of automotive spare parts was largely dependent on conventional retail methods. Customers typically visited multiple automobile spare parts shops to identify suitable products and compare prices. This traditional approach presented several practical challenges and inefficiencies.

Limitations of the Existing System

Offline purchasing process

Time-consuming product search

Limited product availability

- Manual inventory checking
- Lack of centralized product information
- Difficult price comparison
- No online shopping convenience
- Limited customer accessibility
- Absence of digital shopping cart
- No responsive online platform

In many cases, customers were required to depend on shopkeepers for product recommendations, which could result in incorrect product selection due to communication gaps or limited product knowledge.

Traditional stores generally lack modern digital interfaces that allow customers to independently browse available products, compare specifications, and make informed purchasing decisions.

Additionally, businesses relying solely on offline sales have limited opportunities to reach customers outside their local geographical area.

These limitations highlight the necessity of adopting digital solutions capable of providing improved accessibility, product visibility, and customer convenience.

The proposed system is a web-based e-commerce application developed using Python Flask with SQLite as the backend database. The application has been designed to simplify the process of browsing and managing automotive parts while providing customers with an efficient and responsive shopping experience.

Unlike traditional systems, the proposed application organizes automotive products into multiple categories, allowing users to quickly locate the required products through an intuitive navigation structure.

The system provides a responsive interface developed using Bootstrap, ensuring compatibility across desktops, tablets, and mobile devices.

The Shopping Cart module allows users to:

- Add products
- Remove products
- Modify quantities
- View subtotal
- View total amount

Flask Session Management is used to temporarily maintain cart information during user interaction.

SQLite provides a lightweight and efficient database for storing product details while Flask routing enables smooth navigation between different application modules.

The modular architecture adopted in the project allows future expansion without affecting the existing implementation.

Advantages of the Proposed System

The proposed Automotive Store offers several advantages over traditional purchasing methods.

Customer Benefits

Easy product browsing

Organized categories

Attractive interface

Faster purchasing process

Mobile-friendly website

Improved accessibility

Better shopping experience

Technical Benefits

Lightweight Flask framework

SQLite integration

Modular application architecture

Session-based cart management

Responsive Bootstrap interface

Easy maintenance

Future scalability

Business Benefits

Digital product showcase

Wider customer reach

Better product organization

Reduced manual effort

Improved customer satisfaction

Foundation for future e-commerce expansion

The proposed system successfully addresses the limitations of the traditional automotive parts purchasing process while providing a scalable platform that can be enhanced with advanced features such as authentication, payment integration, inventory management, and administrative control in future versions.

TECHNOLOGIES USED

The successful development of the Automotive Store – E-Commerce Website for Automotive Parts & Accessories required the integration of several frontend and backend technologies. Each technology was selected based on its simplicity, reliability, ease of implementation, and suitability for developing a lightweight e-commerce application.

The combination of these technologies enabled the development of a responsive, efficient, and user-friendly web application while providing practical exposure to modern Full Stack Web Development.

Python

Python is a high-level, interpreted programming language widely used in web development, data science, artificial intelligence, automation, and software engineering. It is known for its simple syntax, readability, and extensive collection of libraries.

In this project, Python serves as the primary backend programming language. It is responsible for implementing application logic, routing, session management, product processing, and communication with the SQLite database.

Advantages of Python

- Simple and readable syntax
- Faster application development
- Large community support
- Platform independent
- Extensive standard libraries
- Easy integration with Flask

Flask Framework

Flask is a lightweight Python web framework that simplifies web application development. It provides routing, template rendering, request handling, session management, and database integration while allowing developers to organize projects efficiently.

In the Automotive Store project, Flask is responsible for:

URL Routing

Template Rendering

Session Management

Backend Processing

Database Connectivity

Product Display Logic

Flask follows a modular architecture that makes the application easy to maintain and extend.

Advantages of Flask

Lightweight

Easy to learn

Flexible architecture

Secure session handling

Excellent documentation

Suitable for small and medium-scale projects

HTML5

HTML5 (Hyper Text Markup Language) is the standard markup language used to structure web pages.

In this project, HTML5 is used for creating:

Homepage

Navigation Bar

Category Cards

Product Cards

Shopping Cart Page

Footer

Forms

Layout Structure

HTML provides the basic framework upon which all other frontend technologies operate.

CSS3

CSS3 (Cascading Style Sheets) is used for designing attractive and visually appealing web pages.

The Automotive Store uses CSS for:

Typography

Card Styling

Buttons

Navigation Bar

Footer

Responsive Layout

Hover Effects

Colors

Margins and Padding

Custom CSS enhances the appearance of the website while maintaining consistency throughout the application.

Bootstrap

Bootstrap is an open-source frontend framework that simplifies responsive web design.

Bootstrap provides:

Responsive Grid System

Cards

Navbar

Buttons

Forms

Utility Classes

Mobile Compatibility

SQLite

SQLite is a lightweight relational database management system used to store application data.

In the Automotive Store project, SQLite stores:

Product Information

Category Information

Product Prices

Product Descriptions

Product Images

SQLite was selected because it requires no separate server installation and integrates seamlessly with Flask.

Git & GitHub

Git is a distributed version control system used for tracking changes in software projects.

GitHub provides an online platform for hosting Git repositories and collaborating on software projects.

Although the project repository is yet to be uploaded, GitHub will be used for:

Source Code Management

Version Control

Backup

Collaboration

Project Sharing

GitHub Repository: (To be Updated)

Technology Stack Summary

Technology	Purpose
------------	---------

Python	Backend Programming
--------	---------------------

Flask	Web Framework
-------	---------------

HTML5	Web Page Structure
-------	--------------------

CSS3	Styling
------	---------

Bootstrap	Responsive UI
-----------	---------------

SQLite	Database
--------	----------

Git	Version Control
-----	-----------------

GitHub	Code Repository
--------	-----------------

SOFTWARE REQUIREMENTS

The Automotive Store project was developed using commonly available software tools that support efficient web application development.

Software Requirements Table

Software	Purpose
Windows 10 / Windows 11	Operating System
Visual Studio Code	Source Code Editor
Python 3.x	Programming Language
Flask	Backend Framework
SQLite	Database
HTML5	Frontend Structure
CSS3	Styling
Bootstrap 5	Responsive Design
Git	Version Control
GitHub	Repository Management
Google Chrome / Microsoft Edge	Web Browser
Development IDE	

Visual Studio Code was used as the Integrated Development Environment (IDE).

It provides:

Intelligent Code Completion

Syntax Highlighting

Debugging

Extension Support

Integrated Terminal

Git Integration

Browser Support

The application has been tested on:

Google Chrome

Microsoft Edge

The responsive design allows proper functioning on desktops, laptops, tablets, and smartphones.

HARDWARE REQUIREMENTS

The project does not require specialized hardware and can be executed on any standard computer with moderate specifications.

Hardware Requirements Table

Component	Specification
Processor	Intel Core i3 / AMD Ryzen 3 or Above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	10 GB Free Disk Space
Operating System	Windows 10 / Windows 11
Display	1366 × 768 Resolution or Higher
Internet Connection	Required for Development Resources
Keyboard & Mouse	Standard Input Devices

The lightweight architecture of Flask and SQLite allows the application to perform efficiently without demanding hardware resources.

DEVELOPMENT ENVIRONMENT

The Automotive Store application was developed in a structured software development environment to ensure code organization, maintainability, and efficient debugging.

The development environment consisted of:

Python Virtual Environment (venv)

Flask Framework

SQLite Database

Visual Studio Code

Bootstrap 5

HTML5

CSS3

Git

The project followed a modular directory structure where templates, static resources, database files, and application logic were maintained separately.

Using a virtual environment ensured dependency isolation, preventing package conflicts during development.

Version control practices were followed using Git, while GitHub will be used for repository hosting after project completion.

Using Bootstrap significantly reduced frontend development time while ensuring **the website remains responsive across different devices.**

4.1 Code submission (Github link) : [Aavutisanjay/test_code](https://github.com/Aavutisanjay/test_code)

4.2 Report submission (Github link) :

5 Proposed Design

System Design is one of the most important phases of software development. It provides a blueprint for developing the application by defining the architecture, components, modules, data flow, and interactions among different parts of the system. A well-designed system improves scalability, maintainability, reliability, and performance.

The Automotive Store application has been designed using a modular architecture based on the Flask web framework. The frontend and backend components work together to provide users with a seamless online shopping experience. The system separates presentation, business logic, and database operations, making the application easier to understand, maintain, and extend in the future.

The application consists of the following major components:

User Interface

Flask Backend

SQLite Database

Session Management

Product Categories

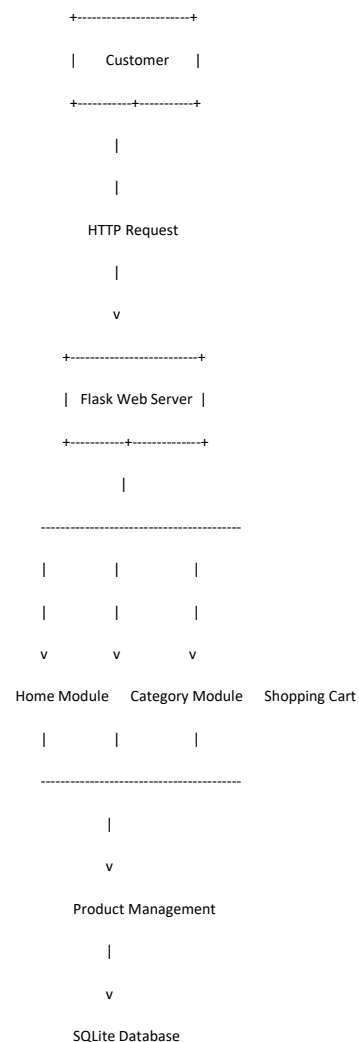
Shopping Cart

Each component communicates efficiently with the others to provide a responsive and user-friendly web application.

5.1 High Level Diagram

The High-Level Architecture represents the interaction between the user, web application, backend, and database.

Figure 16.1 High-Level Architecture



Explanation

The customer accesses the Automotive Store using a web browser. The request is received by the Flask web server, which processes the request and directs it to the appropriate module.

Depending on user interaction, the application retrieves product information from the SQLite database and displays it through HTML templates.

The Shopping Cart module temporarily stores selected products using Flask sessions, allowing users to continue shopping without losing cart information.

5.2 Low Level Diagram

The Low-Level Architecture illustrates how different modules communicate internally within the application.

User



Homepage



Category Selection



Product Listing



Product Details



Add to Cart



Flask Session



Cart Display



Quantity Update



Total Calculation



Checkout (Future Scope)

Explanation

The application begins with the Homepage where users browse available automotive categories.

Selecting a category displays all related products.

Users can add products to the Shopping Cart.

The cart data is maintained using Flask Session Management.

Users can increase or decrease quantities, remove products, and view updated totals.

The Checkout module is planned as a future enhancement.

5.3 Interfaces

The Automotive Store application provides a simple, user-friendly, and responsive interface that enables users to browse automotive products efficiently. The interfaces were developed using HTML5,

CSS3, Bootstrap 5, and Flask templates, ensuring compatibility across desktops, laptops, tablets, and mobile devices.

The major interfaces of the application are described below

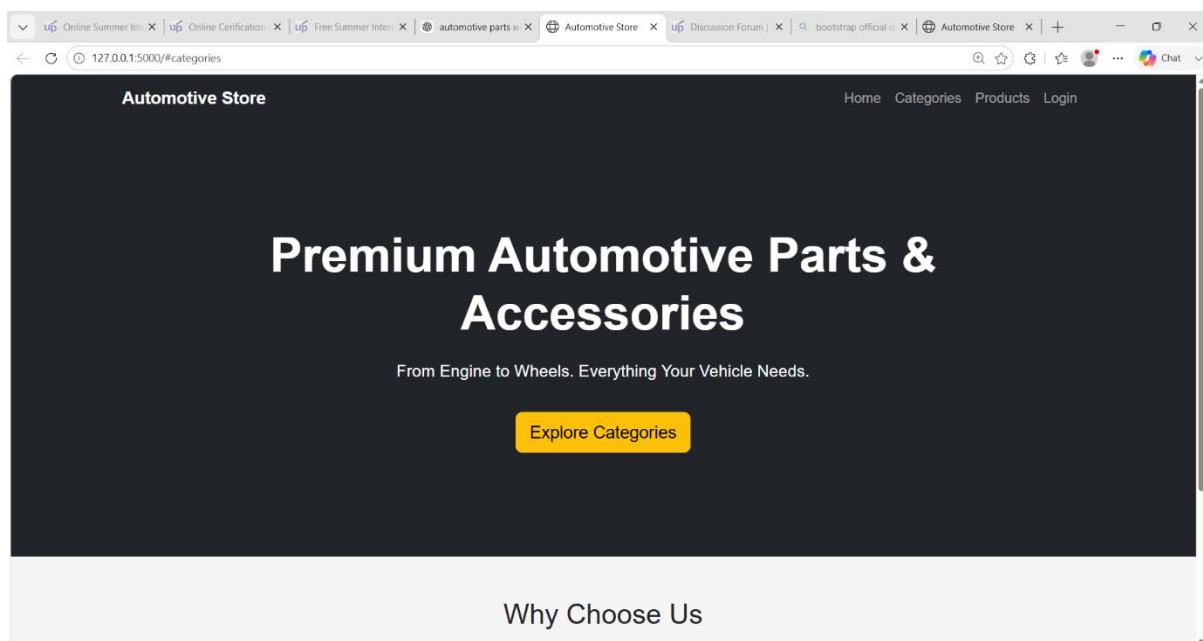
5.3.1 Homepage Interface

The Homepage serves as the entry point of the application. It provides users with quick navigation to different automotive product categories.

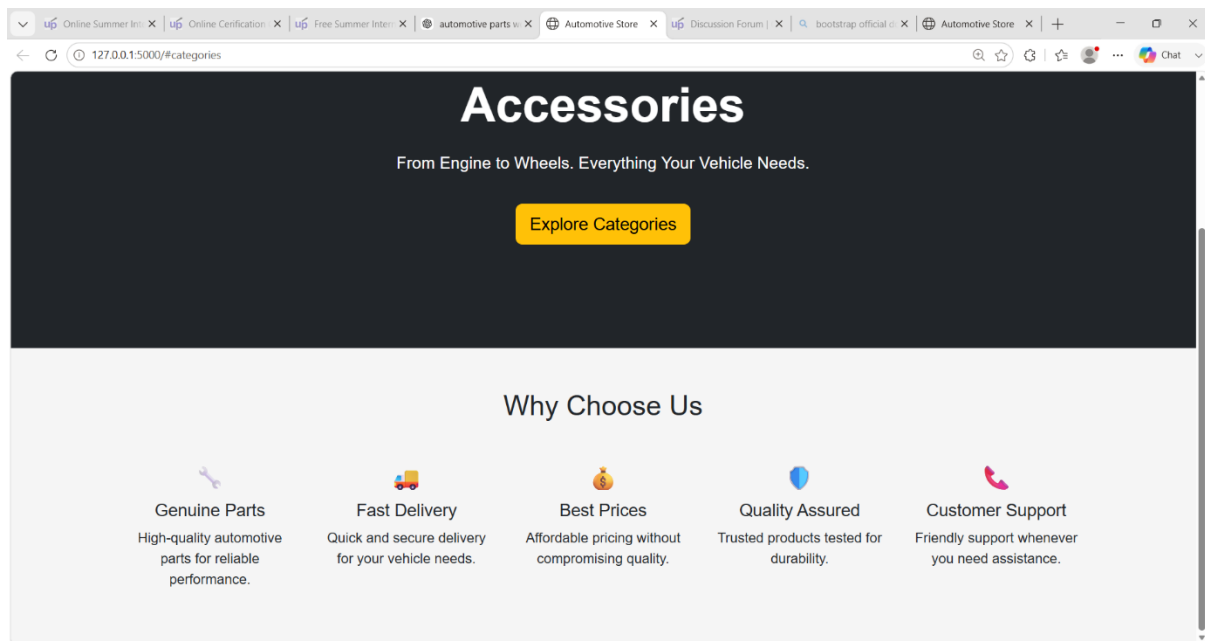
Features:

- Navigation Bar
- Hero Banner
- Product Categories
- Why Choose Us Section
- Footer
- Responsive Layout

Homepage interface figure 5.3.1.1



Homepage interface figure 5.3.1.2



5.3.2 Category Interface

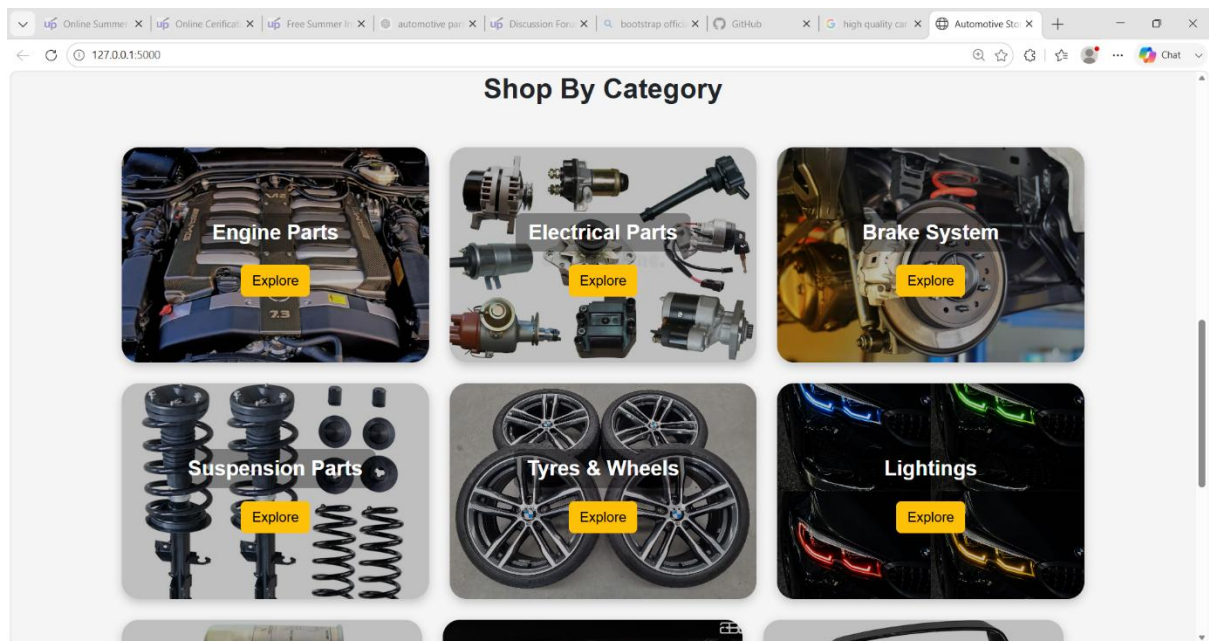
The Category Interface displays automotive products grouped into predefined categories.

The implemented categories include:

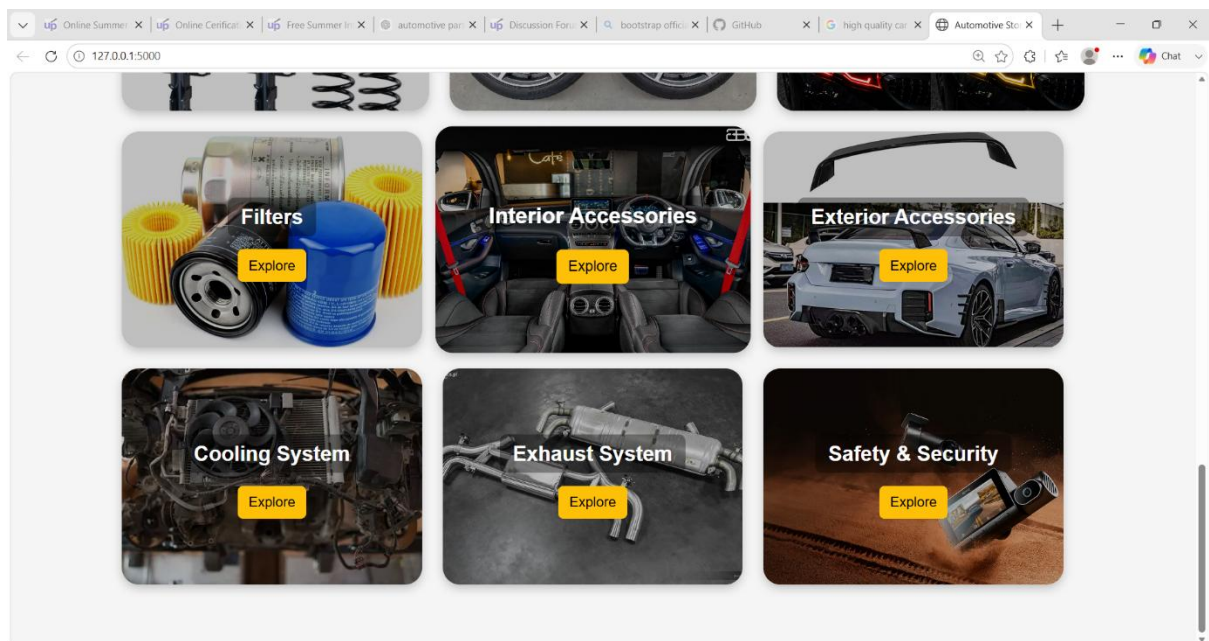
- Engine Parts
- Brake System
- Electrical Parts
- Suspension Parts
- Tyres & Wheels
- Lighting
- Filters
- Interior Accessories
- Exterior Accessories
- Cooling System
- Exhaust System
- Safety & Security

Users can select any category to view the corresponding products.

Category Interface figure 5.3.2.1



Category Interface figure 5.3.2.2



5.3.3 Product Interface

The Product Interface presents complete information about individual products.

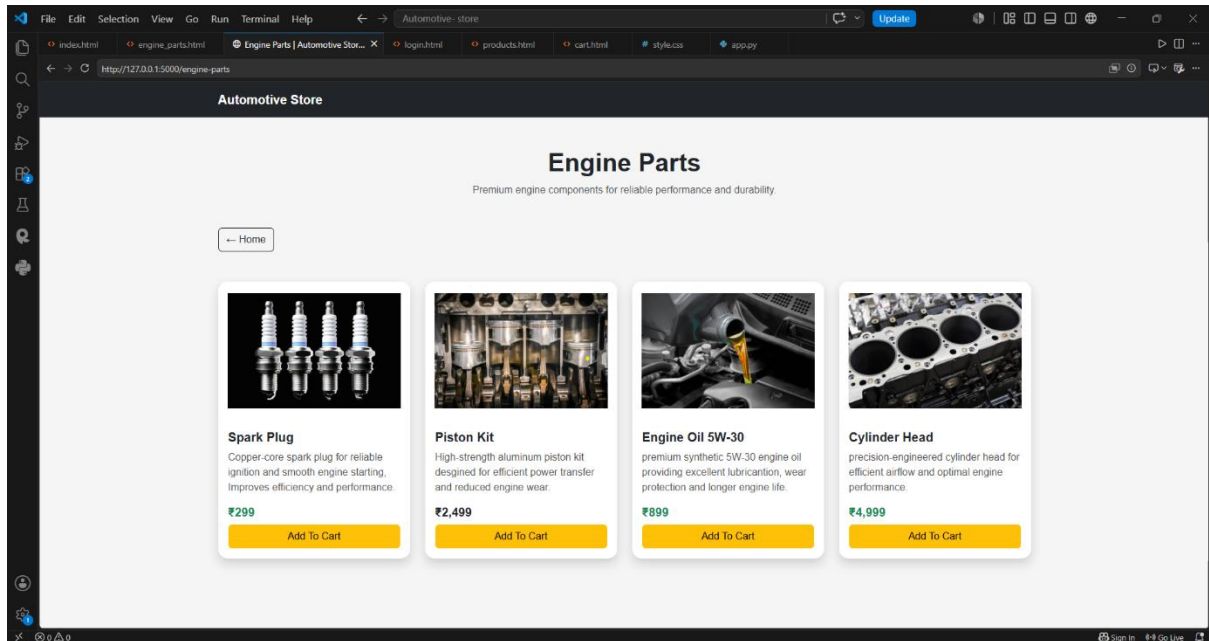
Each product card contains:

- Product Image
- Product Name
- Description
- Price

- Add to Cart Button

The product information is retrieved dynamically from the SQLite database through the Flask backend.

Product interface figure 5.3.3.1



Shopping Cart Interface 5.3.4

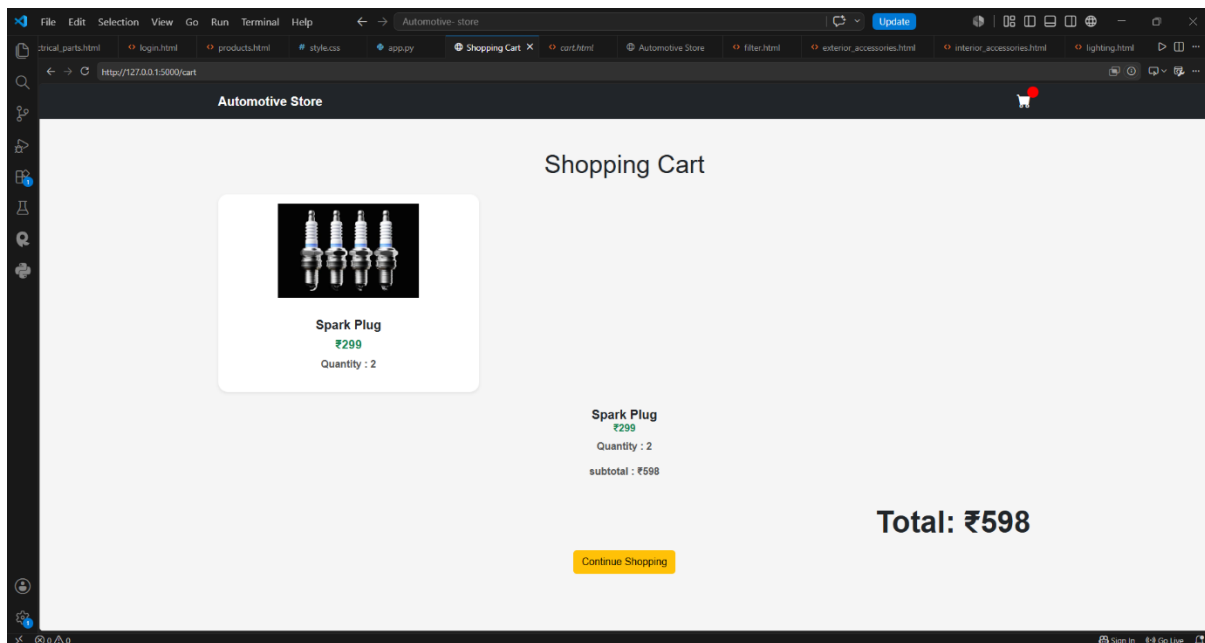
The Shopping Cart Interface enables users to manage selected products before checkout.

Users can:

- Add Products
- Remove Products
- Increase Quantity
- Decrease Quantity
- View Total Amount

Flask Session Management is used to preserve cart information during browsing.

Shopping cart interface figure 5.3.4.1



6 Performance Test

Software testing is an essential phase of the Software Development Life Cycle (SDLC). It ensures that the developed application functions correctly, meets user requirements, and provides a reliable user experience. During the development of the Automotive Store application, various testing methods were performed to verify the correctness of each module.

The objective of testing was to identify defects, validate functional requirements, and ensure smooth interaction between the frontend and backend components.

The application was tested locally using the Flask development server and modern web browsers such as Google Chrome and Microsoft Edge.

6.1 Types of Testing Performed

Unit Testing

Each individual module, such as the Homepage, Product Module, Category Module, and Shopping Cart, was tested independently to verify its functionality.

6.2 Integration Testing

Integration testing was carried out to ensure that all modules communicated correctly with each other. For example, selecting a category correctly displayed products, and adding products updated the shopping cart.

6.3 System Testing

The complete application was tested as a whole to ensure that all functionalities worked together as expected.

6.4 User Interface Testing

The responsiveness and appearance of the website were tested across different screen sizes to ensure compatibility with desktops, laptops, tablets, and mobile devices.

6.5 Test Cases

Test Case ID	Module	Test Scenario	Expected Result	Status
TC-01	Homepage	Open Homepage	Homepage displays correctly	Passed
TC-02	Categories	Select Category	Related products displayed	Passed
TC-03	Products	View Product Details	Product information displayed	Passed
TC-04	Shopping Cart	Add Product	Product added successfully	Passed
TC-05	Shopping Cart	Update Quantity	Quantity updated correctly	Passed
TC-06	Shopping Cart	Remove Product	Product removed successfully	Passed
TC-07	Responsive Design	Open on Mobile	Layout adjusts correctly	Passed
TC-08	Database	Retrieve Products	Product details loaded successfully	Passed.

6.6 RESULTS AND DISCUSSION

The Automotive Store project was successfully designed and developed as a functional web-based e-commerce application. The application provides users with a simple, responsive, and organized platform for browsing automotive parts and accessories.

The project successfully achieved its primary objectives:

Developed a responsive web interface using HTML5, CSS3, and Bootstrap.

Implemented backend functionality using Python and Flask.

Stored and retrieved product information using SQLite.

Implemented a session-based shopping cart.

Organized products into multiple automotive categories.

Provided an intuitive user interface for improved user experience.

The modular architecture adopted during development makes the application easy to maintain and extend with additional features in the future.

The overall performance of the application was satisfactory during testing, with all implemented modules functioning as expected.

6.7 CHALLENGES FACED

During the internship, several practical challenges were encountered while developing the Automotive Store application. These challenges provided valuable learning opportunities and helped improve problem-solving skills.

Some of the major challenges included:

Understanding the Flask project structure.

Managing routing between multiple web pages.

Integrating SQLite with Flask.

Designing a responsive user interface.

Maintaining session-based shopping cart data.

Organizing automotive products into meaningful categories.

Handling static files such as images and CSS.

Debugging application errors and resolving runtime issues.

Ensuring consistent page layouts across different devices.

Each challenge was addressed through research, experimentation, mentor guidance, and continuous testing.

6.8 SKILLS ACQUIRED DURING THE INTERNSHIP

The internship provided practical exposure to modern web application development and significantly enhanced both technical and professional skills.

Technical Skills

Python Programming

Flask Web Framework

HTML5

CSS3

Bootstrap 5

SQLite Database

SQL Fundamentals

Git Basics

Responsive Web Design

Session Management

Database Connectivity

Software Testing

Professional Skills

Problem Solving

Analytical Thinking

Team Collaboration

Time Management

Documentation

Debugging

Communication Skills

Software Development Life Cycle (SDLC)

Project Planning.

These skills will be valuable for future software development projects and professional careers in the IT industry.

7 My Learnings

The Industrial Internship provided me with valuable practical exposure to software development and helped me bridge the gap between academic learning and real-world application development. During the six-week internship, I gained hands-on experience in designing, developing, testing, and documenting a web-based e-commerce application using modern web technologies.

One of my key learnings was understanding how different technologies work together to build a complete full-stack web application. I learned how the frontend developed using **HTML5, CSS3, and Bootstrap** communicates with the backend developed using the **Flask framework**. I also gained knowledge of how data is stored, retrieved, and managed using the **SQLite database**.

This internship strengthened my understanding of several software development concepts, including routing in Flask, template rendering using Jinja2, session management, database connectivity, responsive web design, and modular programming. I also learned how to organize a project using a proper folder structure and follow systematic development practices.

Working on the Automotive Store project improved my debugging and problem-solving skills. During development, I encountered issues related to routing, image management, shopping cart functionality, responsive layouts, and database integration. Resolving these challenges enhanced my analytical thinking and helped me understand the importance of testing and debugging throughout the software development process.

The internship also introduced me to the Software Development Life Cycle (SDLC), where I learned the importance of planning, designing, implementation, testing, documentation, and continuous improvement while developing software applications.

Apart from technical skills, the internship helped me improve several professional skills such as communication, documentation, time management, self-learning, and adapting to new technologies. I also understood the importance of writing clean, maintainable code and developing user-friendly applications.

Overall, this internship has significantly increased my confidence in full-stack web development and provided me with practical knowledge that will be highly beneficial for my future academic projects and professional career in the software industry.

8 FUTURE SCOPE

Although the current version of the Automotive Store application provides essential e-commerce functionality, several advanced features can be implemented in future versions to enhance usability, security, and scalability.

The planned future enhancements include:

User Registration and Login

Secure Authentication

Wishlist Functionality

Admin Dashboard

Product Management Panel

Checkout Module

Online Payment Gateway Integration

Order History

Customer Reviews and Ratings

Product Search and Filters

Inventory Management

Email Notifications

Invoice Generation

Order Tracking

AI-Based Product Recommendations

Implementing these features will transform the application into a complete and scalable e-commerce platform capable of supporting real-world business requirements.

9 CONCLUSION

The Automotive Store – E-Commerce Website for Automotive Parts & Accessories was successfully developed during the Industrial Internship using Python, Flask, HTML5, CSS3, Bootstrap, and SQLite.

The project provided practical experience in designing and developing a web-based application by integrating frontend and backend technologies. Throughout the internship, theoretical knowledge gained during the MCA program was applied to solve real-world development challenges.

The application successfully demonstrates key concepts such as responsive web design, database connectivity, session management, modular programming, and structured software development.

The internship also enhanced technical knowledge, problem-solving abilities, teamwork, documentation skills, and overall confidence in software development practices.

Overall, the internship proved to be an enriching learning experience that strengthened both technical expertise and professional competencies, preparing for future opportunities in the software industry.